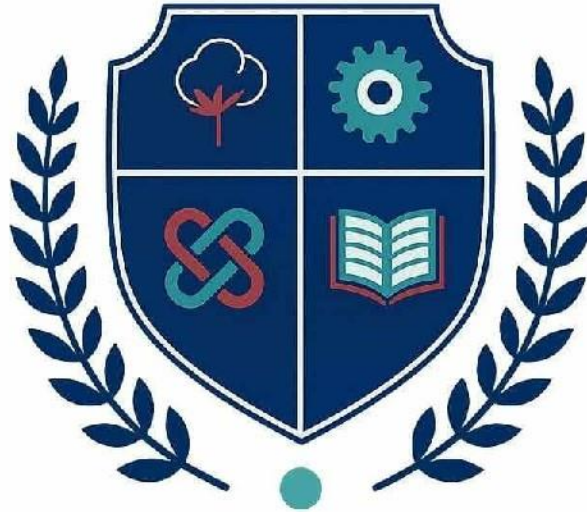






University of Kamalia

PROJECT REPORT

Smart Student Result Management System

COURSE

Programming Fundamentals

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Introduction

The **Smart Student Result Management System** is a Python program that helps manage student results. It allows the user to enter student information and subject marks. The program automatically calculates the total marks, average, percentage, grade, and final result. It also displays the strongest and weakest subject of each student.

Objective

The main objective of this project is to develop a simple Student Result Management System using Python and apply the basic concepts of Programming Fundamentals. This project also helps reduce manual calculations and makes result processing easier.

Features

- Menu using **match-case**
- Add student record
- Enter marks for five subjects
- Validate marks between 0 and 100
- Calculate total marks, average, percentage, and grade
- Display pass or fail result
- Search student by roll number
- View all student records
- Display subject-wise performance
- Show the strongest and weakest subject

Concepts Used

- Variables
- Lists
- User-defined functions
- if-elif-else
- for loop
- while loop
- match-case
- Input validation

Conclusion

This project helped me understand the basic concepts of Python programming. I learned how to use variables, lists, functions, loops, conditions, and match-case in a practical program. It also improved my programming and problem-solving skills.

Screenshots

Figure 1: Main Menu

```
=== SMART STUDENT RESULT MANAGEMENT SYSTEM ===
```

- 1.Add Student
- 2.View Student Result
- 3.View All Students
- 4.Search Student
- 5.Exit

Enter your choice (1-5):

↑↓ for history. Search history with c-↑/c-↓

Figure 2: Add Student

```
Enter your choice (1-5): 1
```

```
== Add Student ==
```

```
Enter Roll Number: 11
```

```
Enter Student Name: Muqaddas Zaheer
```

```
Enter Eng Marks: 77
```

```
Enter Math Marks: 88
```

```
Enter Phy Marks: 90
```

```
Enter Comp Marks: 99
```

```
Enter Urdu Marks: 86
```

```
Student Added Successfully!
```

Figure 3: View Student Result

```
=====
Enter your choice (1-5): 2
Enter Roll Number to Search: 11
=== STUDENT RESULT CARD ===
Roll Number: 11
Student Name: Muqaddas Zaheer
Subject Marks
Eng : 77
Math : 88
Phy : 90
Comp : 99
Urdu : 86
Total Marks: 440 /500
Average: 88.0
Percentage: 88.0 %
Grade: A
Result: Pass
==Subject Performance==
Eng : Good
Math : Excellent
Phy : Excellent
Comp : Excellent
Urdu : Excellent
Strongest Subject: Comp
Weakest Subject: Eng
```

Figure 4: View All Students

```
=====
Enter your choice (1-5): 3
== ALL STUDENTS RECORD ==
Roll Number: 11
Name: Muqaddas Zaheer
Percentage: 88.0 %
Grade: A
Result: Pass
=====
```


Figure 5: Search Student

```
Enter your choice (1-5): 4
Enter Roll Number to Search: 11
=== STUDENT RESULT CARD ===
Roll Number: 11
Student Name: Muqaddas Zaheer
Subject Marks
Eng : 77
Math : 88
Phy : 90
Comp : 99
Urdu : 86
Total Marks: 440 /500
Average: 88.0
Percentage: 88.0 %
Grade: A
Result: Pass
==Subject Performance==
Eng : Good
Math : Excellent
Phy : Excellent
Comp : Excellent
Urdu : Excellent
Strongest Subject: Comp
Weakest Subject: Eng
```

Figure 6: Exit Program

```
=== SMART STUDENT RESULT MANAGEMENT SYSTEM ===
1.Add Student
2.View Student Result
3.View All Students
4.Search Student
5.Exit
Enter your choice (1-5): 5
Thank you for using the system.
Program Ended Successfully.
```